Jaeseung Lee

Flushing, NY | E-Mail: jaeseung.lee124@gmail.com | Mobile: (551) 330-8986

Digital and computational pathology professional with experience in computational pathology research, machine learning model evaluation, and large-scale biomedical dataset management. Skilled in Python, R, Linux, Git, metadata organization, data quality control, and structured research workflows. Experienced in collaborating with pathologists, and bioinformatics researchers to contribute to translational research projects.

EXPERIENCE

DIGITAL PATHOLOGY WEB PORTAL 2026 (Ongoing)

Independent Project | [Demo] [GitHub]

- Built a web-based digital pathology research tool for reviewing whole-slide imaging datasets, including case-level navigation, slide selection, metadata display, and interactive image exploration.
- Implemented a StarDist-based nuclei segmentation and cell counting workflow for user-defined ROIs, enabling automated count estimation and segmentation overlay visualization directly within the web portal.

COLUMBIA UNIVERSITY IRVING MEDICAL CENTER New York, NY

Department of Pathology and Cell Biology

Staff Associate I Nov 2023 – Jun 2024

- Processed single-nucleus RNA-seq datasets from FASTQ files using Cell Ranger; performed downstream quality control, clustering, cell-type annotation, and differential expression analysis; managed structured tracking of raw files, processed outputs, whole-slide images, and sample metadata.
- Developed computational pipelines for spatial transcriptomics analysis, including cell-cell interaction modeling and transcriptomic characterization, achieving up to 1200× speedup through GPU acceleration; co-authored a *Nature Communications* publication and released the code on GitHub. [Paper] [GitHub]
- Evaluated state-of-the-art deep learning models for WSI-based Huntington's disease classification using ROC-AUC, F1-score, and precision/recall; generated attention heatmaps to support model explainability and discovery of potential disease-associated imaging features.

Visiting Researcher Apr 2023 – Jun 2023

- Processed RNA-seq datasets through quality control, alignment, and differential expression analysis; maintained structured tracking of raw files, processed outputs, sample metadata, and processing status.

KOREA UNIVERSITY OF TECHNOLOGY AND EDUCATION South Korea

Department of Computer Engineering

Research Assistant Mar 2021 – Dec 2022

- Developed a machine learning model to detect physiological responses to music using wearable sensor data, achieving an F1-score of 0.81; co-authored a peer-reviewed publication.

SKILLS

- **Programming**: Python, R, C/C++, Bash
- **Tools:** Cell Ranger, Space Ranger, Loupe Browser
- **Digital Pathology:** OpenSlide, QuPath, ImageJ/Fiji
- **R Libraries**: edgeR, DESeq2, Seurat
- **Web Dev**: HTML/CSS, JavaScript, Codex
- **AI/ML:** TensorFlow, PyTorch, Scikit-learn

PUBLICATIONS

- Ma, M., Paryani, F., Jakubiak, K., Xia, S., Antoku, S., Kannan, A., ***Lee, J., …*** Sims, P. A. & Al-Dalahmah, O. *The spatial landscape of glial pathology and T cell response in Parkinson's disease substantia nigra. Nature Communications*. 2025; 16:7146. [Paper] [GitHub]
- Lee, E., Min, C., ***Lee, J.***, Yu, J. and Kang, S. *Groovemeter: Enabling music engagement-aware apps by detecting reactions to daily music listening via Earable sensing. Proc. 31st ACM Int. Conf. on Multimedia*. 2023. [Paper]

EDUCATION

KOREA UNIVERSITY OF TECHNOLOGY AND EDUCATION South Korea

Bachelor of Science in Electrical Engineering Feb 2021